

PROMPT ENGINEERING

MODEL USED: Perplexity

1.Zero shot + Role based:

Result:

Hi [Recipient Name],

I hope you're well. I'm Bhadrিকা from [Your Company], and we've built [Product Name] — an analytics platform that helps [industry/role] teams turn raw data into faster, actionable decisions.

Many teams we work with face three common problems: slow report generation, fragmented data sources, and low confidence in predictions. [Product Name] solves these by:

- Combining data pipelines into one view, reducing time-to-insight.
- Providing automated, explainable analytics dashboards that non-technical users can act on.
- Delivering deployment-ready models and alerts so teams can catch issues earlier.

A quick example: a mid-sized e-commerce client cut decision time by 60% and increased campaign ROI by 18% within three months after adopting [Product Name].

Would you be open to a 20-minute call next week so I can show a tailored demo and explore whether this might help your team? I'm available Tuesday–Thursday, 10:00–15:00 IST, and happy to adapt to your schedule.

Thanks for your time — looking forward to hearing from you.

Best

Bhadrিকা

[Title],

Email: [your email] | Phone: [phone] | Website: [url]

[Your

regards,

Murugan

Company]

PROMPT WITH ROLE:

You are a senior sales rep who has closed 200+ enterprise deals. You know that cold emails fail when they talk only about features. They work when they name a specific pain point, Write a cold email to the VP of Marketing about our analytics software.

Hi [VP Name],

I'll be direct — most marketing teams I meet lose weeks and budget because they can't confidently answer one question: which campaigns actually move revenue, not just clicks.

At [Your Company], our analytics platform ties campaign activity to real revenue signals (LTV, churn, and pipeline velocity) and surfaces the single campaign or channel adjustments that consistently improve ROI. Clients say it removes the guesswork that creates these three costly issues:

- Misallocated spend across channels that drive impressions but not conversions.
- Long decision cycles waiting on stitched-together reports from multiple tools.
- Overconfidence in metrics (CTR, impressions) that don't predict revenue lift.

Example: a mid-market SaaS customer shifted 28% of their paid budget to two underused channels after our insight and saw a 22% increase in MQL-to-SQL conversion rate within eight weeks.

If you want, I can send a 7-slide, tailor-made dashboard for your current top campaign in 48 hours — no demo needed — so you can see which budget moves would most likely increase revenue. Would Thursday or Friday work for me to send that over?

Best,

Bhadrika

Senior

Sales,

[Your

Murugan

Company]

Email: [your email] | Phone: [phone] | Website: [url]

2.Few shot+ Structured:

PROMPT 1: Feature Update (Sleep Tracking) Subject: We watched you sleep. In a cool way. PulseX now tracks your sleep stages — light, deep, REM — so you can finally stop guessing why you feel like garbage at 2pm. Here's how it works: wear your band to bed. Wake up. Open the app. See your sleep score. Judge yourself accordingly. It also tracks how often you woke up, how long it took you to fall asleep, and whether your "8 hours" was actually 5 hours of tossing and 3 hours of real rest. No setup needed. The update's already live. Just open the app and tap Sleep. Fair warning — the data might hurt your feelings. [Check your sleep score →]

Subject: We watched you sleep. (In a totally science-y way.)

Hi there,

PulseX now tracks your sleep stages — light, deep, and REM — so you can stop guessing why you feel like garbage at 2 PM.

How it works

- Wear your band to bed. Wake up. Open the app.
- See your sleep score and breakdown: time in light, deep, and REM sleep.
- View wake counts, time to fall asleep, and whether your “8 hours” was really 5 hours of tossing and 3 hours of rest.

No setup required — the update is live. Just open PulseX and tap Sleep.

Heads up: the truth might sting. But it also gives you clear steps to improve your rest.

[Check your sleep score →]

PROMPT 2: Sale Announcement (End of Season) Subject: 30% off. No catch. Okay, one catch. The catch is it ends Sunday. Everything in the store — bands, chargers, straps, the works — 30% off. We'd say "while stocks last" but honestly we have plenty. We just want you to feel urgency. Is it working? If you've been thinking about upgrading your strap or grabbing a second charger for the office, this is the moment. If you haven't been thinking about it, maybe start. We're not going to beg. But we're also not running this sale again until we feel like it. Code: PULSE30 at checkout. Ends Sunday 11:59pm. [Shop the sale →]

Hi,

We're running an end-of-season sale: everything in the store — bands, chargers, straps, and accessories — is 30% off. No hidden conditions. Just use code PULSE30 at checkout.

Key details

- Discount: 30% off sitewide.
- Applies to: all products (bands, chargers, straps, accessories).
- Promo code: PULSE30.
- Ends: Sunday at 11:59 PM (local time).

If you've been meaning to upgrade a strap or add a spare charger for the office, now is a convenient time. Inventory is healthy, but the discount ends on Sunday and won't be repeated immediately.

Shop the sale: [Shop the sale →]

Best,

The PulseX Team

CRAFT PROMPT: <Context> I am currently designing a production grade pipeline using kafka + DBT + Snowflake and terraform integration. My team is full of high quality engineers, and hence I want to make sure, that the code scripted by me too stands out and abides by all the software engineering principles. I want this code to be scalable and optimized in the best possible way.

Role: Act as Senior software engineer with 2 decades of experience, who has worked on high quality enterprise-level scalable applications and is adept with data engineering knowledge and impeccable programming skills. You have worked on a variety of streams like data engineering, machine learning, database engineering. You have unique way for explaining complex concepts in a simpler manner, and also reviewing code in the best possible manner and suggesting the most potent additions to the code which enhances the application quality.

<Action>: I want you to review the code and suggest me where I the code be engineered in a better way, where can it be optimized, which part of the code seems unnecessary and where can i improve better. Which are the strengths and weaknesses in the code, and also suggest me the write versions of the code, after suggesting the changes.

<Format: > Output should be strictly in markdown format. Only if required, perform comparative analysis in tabular format. **Constraints:** The code review should be strictly with the code uploaded. Avoid explaining too technically, rather it should be simpler to follow and implement. Be precise and verbose at the same time.

I'm ready to review it, but I don't actually have your code file in the workspace yet. The only files available right now are the skill/reference files, not your Kafka + dbt + Snowflake + Terraform codebase.

What I need

Please upload the code or paste the relevant files here so I can do a real review based strictly on your implementation.

Best options:

- A zip of the project.
- Individual files like main.tf, variables.tf, Kafka producer/consumer code, dbt models, dbt macros, Snowflake SQL, CI/CD config.
- Or a folder tree plus the most important files.

What I'll review

Once you upload it, I'll give you a markdown-only review with:

- Strengths in the current code.

- Weaknesses and risk areas.
- What is unnecessary or overly complex.
- Where scalability, readability, reliability, and performance can be improved.
- Cleaner rewritten versions of the code sections that need fixing.

Best file set

For this stack, the most useful files are:

- Terraform: providers, modules, variables, outputs, state/backend config.
- Kafka: producers, consumers, schema handling, retry/error logic.
- dbt: models/, macros/, sources.yml, schema.yml, tests, incremental models.
- Snowflake: DDL, task/stream usage, warehouse settings, role/grant setup.
- Deployment/config: env handling, secrets strategy, CI pipeline, linting/formatting configs.

Review format

I'll keep it simple, precise, and practical, and I'll structure it like this:

1. High-level review.
2. Strengths.
3. Issues by priority.
4. Suggested improvements.
5. Rewritten code snippets.
6. Final engineering recommendations.

Upload the codebase or paste the files you want reviewed first.

TRANSCRIPT BEGINS [00:00] Priya Sharma: Alright, let's get started. Nikhil's at the board meeting so he won't be joining, but he wants an update afterward, especially on the timeline. So let's make sure we're clear on where things stand. Arjun, do you want to kick off with the model update?

[00:32] Arjun Mehta: Sure. So the good news is the extraction model is performing significantly better after the retraining last week. We switched from the fine-tuned GPT-3.5 to a Mistral 7B base with LoRA adapters, and the field-level accuracy on invoices went from 78% to 91%. That's on our internal test set of 2,200 documents.

Priya Sharma: That's a big jump. Is that across all document types or just invoices?

Arjun Mehta: Just invoices for now. Purchase orders are at about 84%, which is up from 72%, so also a solid improvement. But contracts are still a problem. We're hovering around 67% on the contract extraction, and honestly I'm not sure fine-tuning alone is going to get us where we need to be on those. [01:45]

Rohit Desai: Yeah, the contract issue is structural. Contracts have way more variability in layout — different firms, different jurisdictions, different formatting conventions. We've been talking about a two-stage approach where we first classify the contract type and then route it to a specialized extraction head, but that adds a lot of complexity to the pipeline.

Priya Sharma: How much complexity are we talking? Another two weeks? A month?

Rohit Desai: Honestly, if we want to do it properly, probably three to four weeks of engineering time. And that's just the model work — Sneha's team would need to build out the routing logic on the data pipeline side too. [02:48]

Sneha Iyer: Hold on, let me push back on that a little. The routing logic isn't the hard part. We can have that up in a few days, it's basically a classifier output feeding into an Airflow branch operator. The hard part is the training data. Do we even have enough labeled contracts to train specialized extraction heads? Last I checked we had maybe 400 labeled contracts total, and that's across all types.

Arjun Mehta: We have 430, to be exact. And if we split by type — NDAs, service agreements, licensing agreements — the smallest bucket is licensing with only 67 examples. That's not enough to fine-tune anything reliable.

Priya Sharma: So what's the alternative? We can't ship with 67% accuracy on contracts. That's the one document type our enterprise clients actually care about the most.

[04:12]

Arjun Mehta: I think the realistic options are: one, we invest in a labeling sprint and get the training data up to at least 200 per contract type before we attempt the two-stage approach. Two, we use a prompted approach for contracts instead of fine-tuning — basically long-context extraction with Claude or GPT-4 using carefully engineered prompts. Three, we descope contracts from the V1 launch and add them in V1.1.

Kavya Nair: Option three is not going to fly with the sales team. I've already got two enterprise prospects — Reliance Digital and Tata Communications — who specifically asked about contract extraction in their last calls. If we pull that from V1, we might lose both deals.

Priya Sharma: Agreed, descope is a last resort. What about option two — the prompted approach? Arjun, is that viable?

[05:30]

Arjun Mehta: It's viable but it changes our cost structure. Right now the fine-tuned Mistral model runs on our own GPU cluster — inference cost is basically zero after the initial hardware investment. If we switch contracts to a prompted approach with an external API, we're looking at roughly 12 to 15 paisa per page, which adds up fast for contract-heavy clients.

Rohit Desai: It's also slower. Fine-tuned model does extraction in about 1.2 seconds per page. A prompted approach with Claude is more like 6 to 8 seconds depending on contract length. For a 40-page contract, that's over five minutes.

Kavya Nair: Is five minutes a dealbreaker though? These aren't real-time use cases. A client uploads a contract and gets the extraction back in five minutes — that's still way faster than their current process, which is someone manually reading the thing for an hour.

[06:52]

Sneha Iyer: I'm less worried about the latency and more worried about the reliability. If we're calling an external API for a core feature, we need proper retry logic, fallback handling, rate limiting — all of that. And we need to figure out the data residency question. Some of these enterprise clients have strict requirements about where their data gets processed. Are we okay sending their contracts to Anthropic's API?

Priya Sharma: That's a good point. Kavya, do we know the data residency requirements for Reliance and Tata?

Kavya Nair: Reliance definitely has a data residency clause — everything has to stay within India. I'd need to check the specifics for Tata. Let me follow up with their procurement team and get clarity this week.

Priya Sharma: Okay, please do that by Monday. We can't make a technical decision without knowing the constraints.

[08:15]

Priya Sharma: Alright, here's what I'm thinking. Let's pursue option one and two in parallel. Arjun, can you and Rohit start the labeling sprint for contracts? Get the annotation team to prioritize contract labeling — I want at least 200 labeled examples per contract type within two weeks.

Arjun Mehta: That's aggressive. The annotation team is currently finishing the invoice validation batch. We'd need to pull them off that.

Priya Sharma: The invoice model is already at 91%. We can pause the invoice annotations. Contracts are the bottleneck.

Arjun Mehta: Fair enough. I'll talk to Meera on the annotation team today and get them redirected.

[09:35]

Priya Sharma: Good. And simultaneously, Rohit, can you prototype the prompted approach? I want to see what accuracy we can get with a well-engineered prompt on the 430 contracts we already have. Give me a benchmark by end of next week.

Rohit Desai: I can do the benchmark, but end of next week is tight. I'm also supposed to be fixing the table extraction bug that's been blocking the QA team.

Sneha Iyer: That table extraction bug is actually causing issues on my side too. The downstream ETL pipeline breaks every time it encounters a document with malformed table outputs. We've had 14 pipeline failures this week alone because of it.

Priya Sharma: Okay, how severe is the table bug? Is it a 'things crash' problem or a 'things are slightly wrong' problem? Sneha Iyer: Things crash. The pipeline fails, the document gets sent to a dead letter queue, and we have to manually reprocess. It's costing us about two hours of engineering time per day just babysitting it.

[11:20]

Priya Sharma: That's not acceptable. Rohit, fix the table bug first. That's priority one. Then start the contract prompt benchmark. Can you have the bug fixed by Monday?

Rohit Desai: I think so, yeah. I've already identified the root cause — it's a parsing issue with nested tables. The model outputs valid JSON but the parser doesn't handle the nesting correctly. Should be a day of work, maybe two.

Priya Sharma: Good. Fix it by Monday, then start the contract benchmark. Get me results by the Friday after — March 27th.

Rohit Desai: Got it.

[12:30]

Kavya Nair: One more thing on the timeline. Our V1 target launch date is April 18th. That's five weeks from now. Are we still tracking for that? Because if the contract approach takes four more weeks of engineering after the benchmark results come in, we're already past the launch date.

Arjun Mehta: Honestly? I think April 18th is at risk. Even if the prompted approach works, integrating it properly with fallback handling, monitoring, and cost tracking is at least two weeks of work. And that's after we know which approach we're going with.

Priya Sharma: I hear you, but I'm not ready to move the date yet. Let's see where we are after the benchmark results on March 27th. If we need to adjust, we'll adjust then. But I don't want us to preemptively relax the timeline. Sometimes deadlines make magic happen.

Kavya Nair: Fair, but can we at least flag it to Nikhil? He's going to ask. Priya Sharma: I'll talk to Nikhil. Let me frame it properly — we're on track for invoices and purchase orders, contracts are the open question, and we'll have clarity by end of month. [14:45]

Priya Sharma: Okay, let's move to the data pipeline update. Sneha, where are we on the ingestion service?

Sneha Iyer: Good progress, actually. The async ingestion pipeline is fully built and deployed to staging. We're processing about 1,200 documents per hour in testing, which is well above our target of 800. The S3-to-processing queue is stable, Airflow orchestration is clean, and we've got proper dead letter queues for failed documents.

Sneha Iyer: The one outstanding issue — apart from the table bug that Rohit mentioned — is the OCR pre-processing for scanned documents. We're using Tesseract right now, and the quality is... not great on low-resolution scans. About 22% of our test documents are scanned, and for those, the extraction accuracy drops by 15 to 20 points because the OCR introduces errors before the model even sees the text.

[16:30]

Arjun Mehta: Have you looked at Azure Document Intelligence? Their OCR is significantly better than Tesseract, especially on degraded scans. We used it at my last company and it was night and day.

Sneha Iyer: I've looked at it. The accuracy is better, no question. But it's another external API dependency, another cost line, and another data residency conversation.

Kavya Nair: Also, sorry to keep bringing this up, but Azure has an India region, right? That might actually solve the data residency issue for both the OCR and the contract extraction if we go with a prompted approach through Azure OpenAI.

Arjun Mehta: Azure OpenAI doesn't have the latest Claude models though. If we're going prompted, Claude is performing noticeably better than GPT-4 on our extraction benchmarks. Like, 6 to 8 points better on contracts specifically.

[18:10]

Priya Sharma: Okay, we're going in circles a bit. Let me try to untangle this. Sneha, can you run a quick comparison of Tesseract versus Azure Document Intelligence this

week? Use 50 of the worst scanned documents. I just need to know if the accuracy improvement justifies adding another vendor dependency.

Sneha Iyer: Yeah, I can have that done by Wednesday.

Priya Sharma: Great. And the data residency question — Kavya is going to find out the requirements from Reliance and Tata by Monday. Once we have that, a lot of these API decisions become clearer. Let's not go back and forth on Azure versus Anthropic until we actually know what's allowed.

Kavya Nair: Makes sense.

[19:45]

Priya Sharma: Anything else before we wrap? We're almost at time. Rohit Desai: One quick thing — the GPU cluster is at 87% utilization. When we scale the annotation team and start running more training jobs for the contract models, we're going to hit capacity. Should I file a request for additional GPU allocation now, or wait?

Arjun Mehta: File it now. The procurement process takes two weeks minimum. Better to have it in the queue and cancel than to need it and not have it.

Priya Sharma: Agreed. Rohit, submit the GPU request today. Estimate what we'd need for the contract training runs and add a 30% buffer.

Rohit Desai: Will do.

[20:55]

Kavya Nair: Oh wait, one more thing. The product demo for the Reliance team is scheduled for April 3rd. That's three weeks out. What are we showing them? Because if contract extraction isn't ready, we need to frame the demo carefully.

Priya Sharma: Good catch. Let's show invoices and purchase orders — those are solid. For contracts, we can show a "coming soon" preview or a limited demo with NDAs only, since those are our strongest contract subtype. Kavya, can you draft a demo flow by end of next week and share it with the team?

Kavya Nair: Sure, I'll put together a demo script and share it by Friday the 20th.
[22:10]

Priya Sharma: Alright. Let me do a quick recap. Arjun — redirect the annotation team to contract labeling, target 200 per type in two weeks. Rohit — fix the table extraction bug by Monday, then start the prompted contract benchmark, results by March 27th. Also submit the GPU request today. Sneha — run the Tesseract versus Azure OCR comparison by Wednesday. Kavya — get data residency requirements from Reliance and Tata by Monday, and draft the demo flow by the 20th. I'll update Nikhil on the timeline situation. Everyone clear?

Arjun Mehta: Clear.

Sneha Iyer: Got it.

Rohit Desai: Yep.

Kavya Nair: All good. Priya Sharma: Great. Good progress on the model side — 91% on invoices is a real milestone. Let’s keep that momentum going. Talk next week.
TRANSCRIPT ENDS — 22:48

Weekly AI Pipeline Sync — Action Items & Decisions Summary

Date: Thursday, March 12, 2026 | **Duration:** 47 min | **Launch Target:** April 18, 2026 (5 weeks away)

Executive Summary

- **Model performance:** Invoice extraction jumped from 78% → **91%** (Mistral 7B + LoRA); POs at **84%** (up from 72%); **contracts stalled at 67%** — the critical enterprise blocker
- **Critical bug:** Nested table parsing breaks ETL pipeline — **14 failures this week**, ~2 hours/day engineering overhead
- **GPU capacity:** Cluster at **87% utilization**; procurement request submitted today
- **V1 launch:** At risk due to contract extraction uncertainty; clarity expected **March 27** after benchmark

Key Decisions Made

Decision	Rationale	Owner	Deadline
Pause invoice annotations; redirect to contract labeling	Invoices at 91% (stable); contracts are bottleneck	Arjun Mehta	Immediate
Fix table extraction bug BEFORE contract benchmark	Pipeline crashes daily; blocks QA & ETL	Rohit Desai	Monday, March 17

Decision	Rationale	Owner	Deadline
Pursue labeling sprint + prompted approach in parallel	+ 67% accuracy unacceptable; need data + benchmark before committing	Arjun Rohit	+ March 27
GPU request submitted now with 30% buffer	Procurement takes 2 weeks; better early than late	Rohit Desai	March 12 (today)
Demo for Reliance (April 3): invoices + POs + NDA preview	Contracts not ready; NDAs are strongest contract subtype	Kavya Nair	April 3

Critical Blockers

1. Contract Extraction (67% accuracy) — V1 Showstopper

Problem: Only 430 labeled contracts total; smallest bucket (licensing) has **67 examples** — insufficient for fine-tuning

Options being pursued:

Approach	Pros	Cons	Timeline
Labeling sprint (200/type)	Builds internal model; zero API cost	Needs 2 weeks; annotation capacity	team 2 weeks
Prompted extraction (Claude/GPT-4)	Fast prototype; better accuracy (6–8 pts vs GPT-4)	₹0.12–0.15/page; 6–8s/page; residency risk	Benchmark by March 27
Descope contracts	Meets April 18 date	Lose Reliance + Tata deals	N/A

Data residency constraint: Reliance requires **India-only processing**; Tata pending confirmation

2. Table Extraction Bug — Pipeline Stability

Root cause: Parser doesn't handle nested JSON tables

Impact:

- 14 pipeline failures this week

- 2 hours/day engineering babysitting
- Documents sent to dead letter queue; manual reprocessing required

Fix timeline: 1–2 days; priority #1 before contract work

3. OCR Quality on Scanned Documents

Problem: 22% of test docs are scanned; Tesseract accuracy drops **15–20 points** on low-res scans

Comparison test: Tesseract vs Azure Document Intelligence (50 worst scans) | **Owner:** Sneha Iyer | **Deadline:** Wednesday, March 19

Action Items (Deadline-Ordered)

This Week (March 12–19)

Action	Owner	Deadline	Status
Submit GPU cluster request (estimate + 30% buffer)	Rohit Desai	Today (March 12)	 Pending
Redirect annotation team to contract labeling	Arjun Mehta	Today	 Pending
Get data residency requirements from Reliance & Tata	Kavya Nair	Monday, March 17	 Pending
Fix nested table parsing bug	Rohit Desai	Monday, March 17	 Critical
Tesseract vs Azure OCR comparison (50 scans)	Sneha Iyer	Wednesday, March 19	 Pending

Next Week (March 20–27)

Action	Owner	Deadline	Status
Draft Reliance demo flow (invoices + POs + NDA preview)	Kavya Nair	Friday, March 20	 Pending
Prompted contract extraction benchmark results	Rohit Desai	Friday, March 27	 Pending

Action	Owner	Deadline	Status
Labeling sprint: 200 contracts/type (4 types)	Arjun Meera	+ ~March 26	 Pending

Follow-ups

Action	Owner	Deadline
Update CTO (Nikhil) on timeline risk	Priya Sharma	After March 27
Check Tata data residency with procurement	Kavya Nair	Monday, March 17

Performance Metrics

Metric	Current	Target	Status
Invoice extraction accuracy	91%	90%	 Exceeded
PO extraction accuracy	84%	80%	 Exceeded
Contract extraction accuracy	67%	90%+	 Blocker
Pipeline throughput	1,200 docs/hr	800 docs/hr	 Exceeded
GPU cluster utilization	87%	<80%	 Near capacity
Scanned doc accuracy drop	-15–20 pts	<5 pts	 Needs OCR fix

Risk Assessment

Risk	Probability	Impact	Mitigation
April 18 launch slip	High	Critical	Benchmark results March 27 will inform decision
Data residency violation	Medium	Critical	Kavya confirming Monday; Azure India region may solve
Reliance/Tata deal loss	Medium	Critical	Contract extraction in V1 is non-negotiable

Risk		Probability	Impact	Mitigation
GPU exhaustion	capacity	High	High	Request submitted; 2-week procurement lead time
External API overrun	cost	Medium	Medium	₹0.12–0.15/page; need cost modeling for contract-heavy clients

Next Meeting

When: Next week (week of March 17)
Focus: Review table bug fix, OCR comparison results, data residency clarity, and GPU request status

Prepared from meeting transcript recorded March 12, 2026, 10:30 AM IST (47 min)